CS & IT ENGINEERING

Digital Logic

Sequential Circuit

Lecture No. [redacted] 11

By- Aditya sir

Recap of Previous Lecture

Topic

Asynchronous

↳ Up/Down

Topics to be Covered

Topic

Sequential Circuit

- Asynchronous
- Synchronous
  - Ring
  - Johnson

About Aditya Jain sir

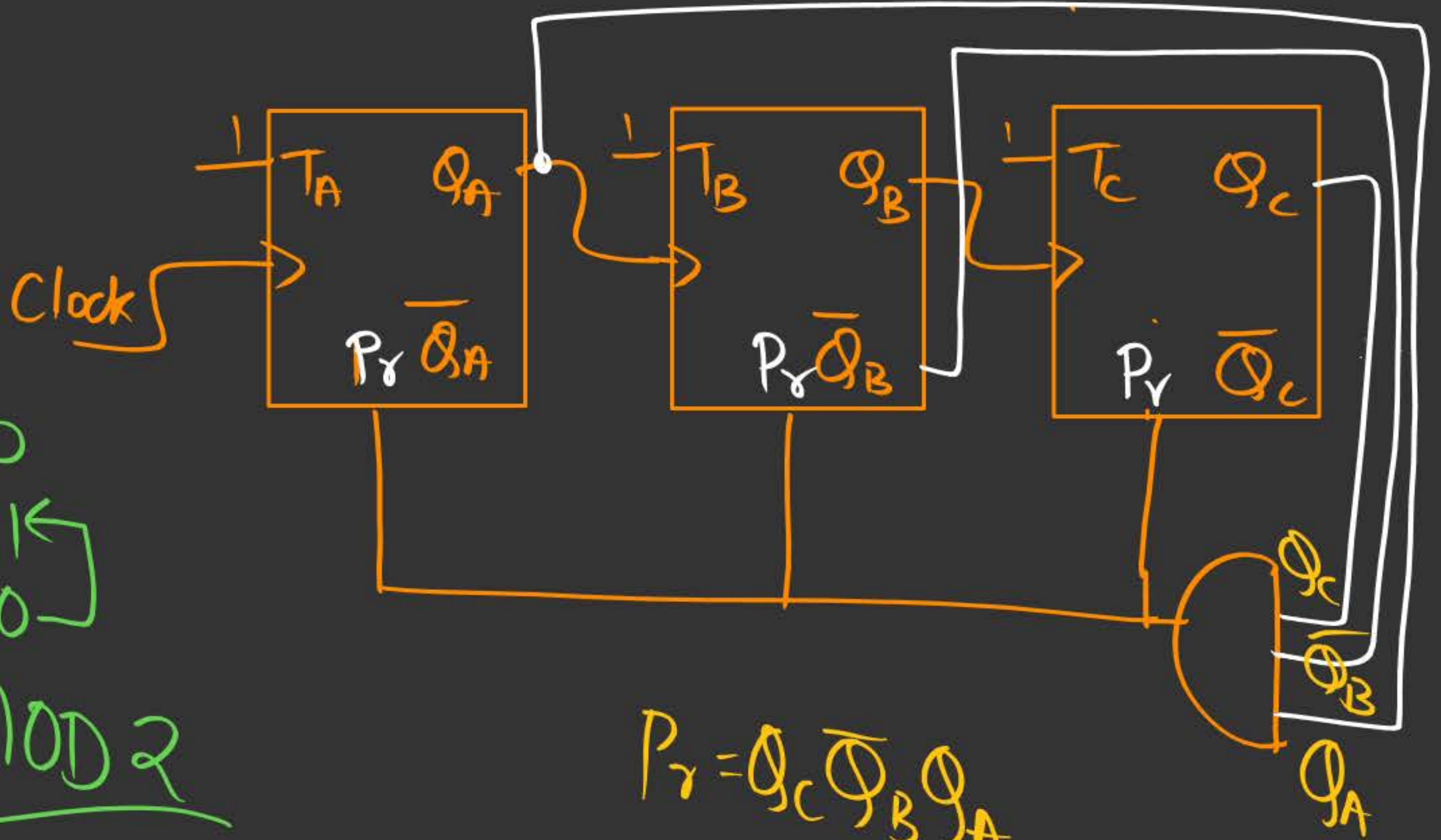
1. Appeared for GATE during BTech and secured AIR 60 in GATE in very first attempt - City topper
2. Represented college as the first Google DSC Ambassador.
3. The only student from the batch to secure an internship at Amazon. (9+ CGPA)
4. Had offer from IIT Bombay and IISc Bangalore to join the Masters program
5. Joined IIT Bombay for my 2 year Masters program, specialization in Data Science
6. Published multiple research papers in well known conferences along with the team
7. Received the prestigious excellence in Research award from IIT Bombay for my Masters thesis
8. Completed my Masters with an overall GPA of 9.36/10
9. Joined Dream11 as a Data Scientist
10. Have mentored 12,000+ students & working professions in field of Data Science and Analytics
11. Have been mentoring & teaching GATE aspirants to secure a great rank in limited time
12. Have got around 27.5K followers on Linkedin where I share my insights and guide students and professionals.

@ADITYASIR_PW

Telegram Link for Aditya Jain sir: https://t.me/AdityaSir_PW

(Q.1) Counter?

Down Ripple Counter



$$P_r = Q_C \bar{Q}_B Q_A$$

000
111
110

MOD 2

$Q_C \bar{Q}_B Q_A$

Clock	Q_C	Q_B	Q_A	P_r
0	0	0	0	0
1	1	1	1	0
2	1	1	0	0
3	1 $\cancel{1}$	1 $\cancel{0}$	$\cancel{1}$ 1	$\cancel{1}$ 0
4	1	1	0	0
5	1 $\cancel{1}$	1 $\cancel{0}$	$\cancel{1}$ 1	$\cancel{1}$ 0
6				
7				
8				
9				

(d) Design a MOD-9 Down Ripple Counter

Telegram

Up = full − down

4 FF

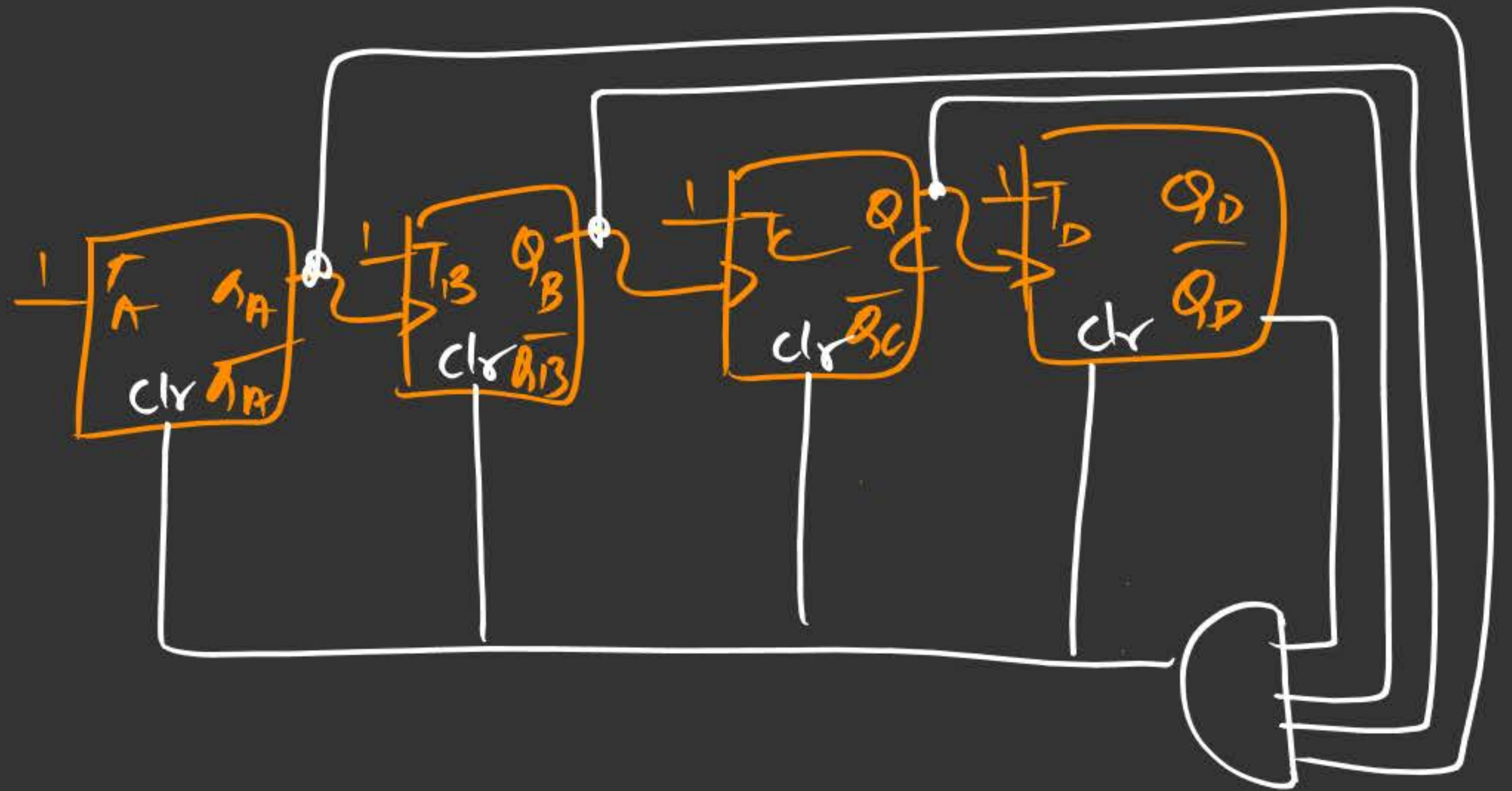
Full = 2^4 = 16

Up = 16 − 9

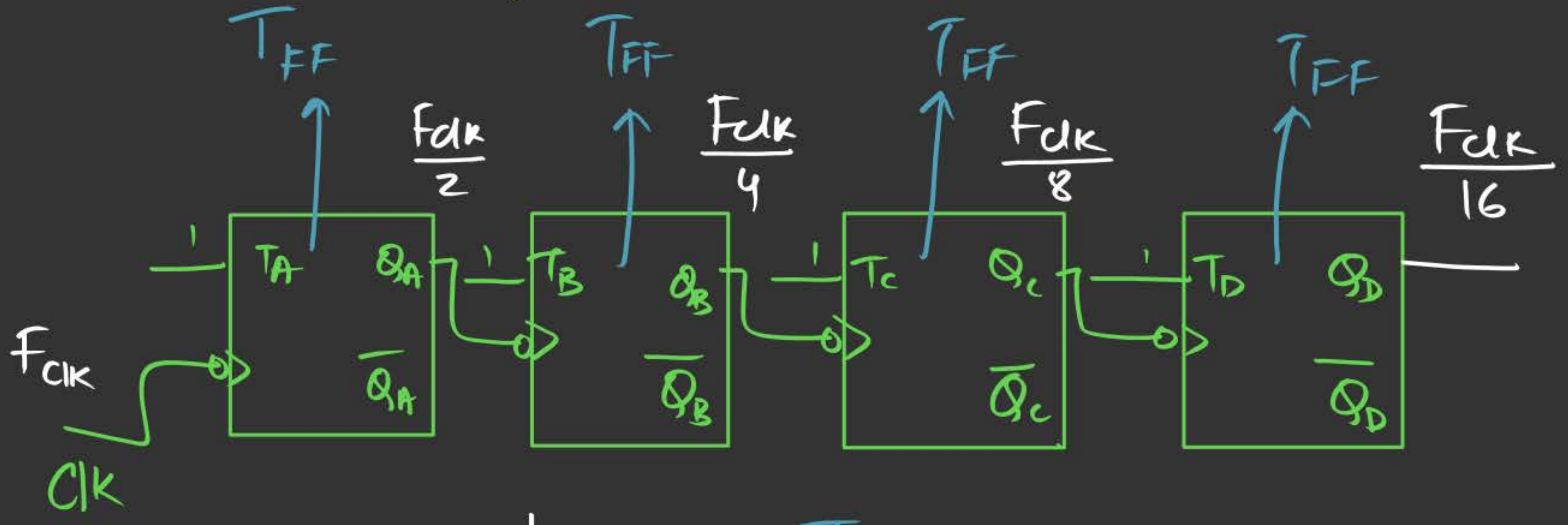
$\boxed{Up = 7}$

$\bar{Q}_D Q_C Q_B Q_A$

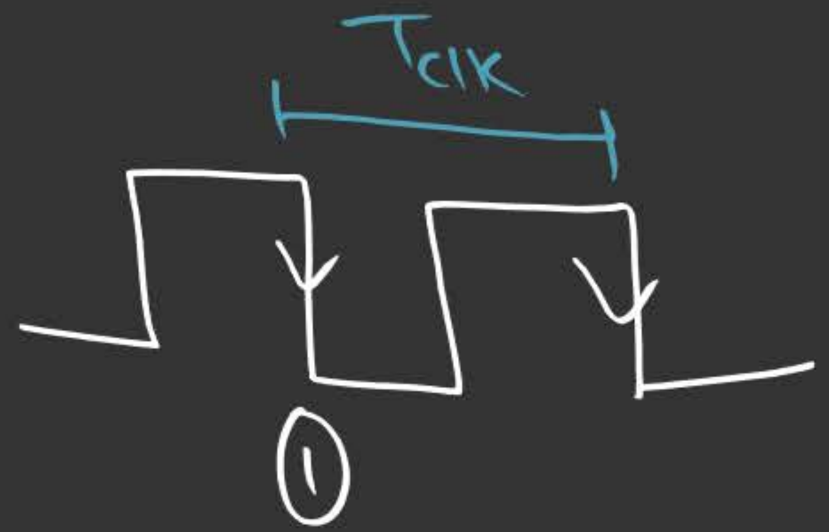
0111



Asynchronous (Ripple Counter)

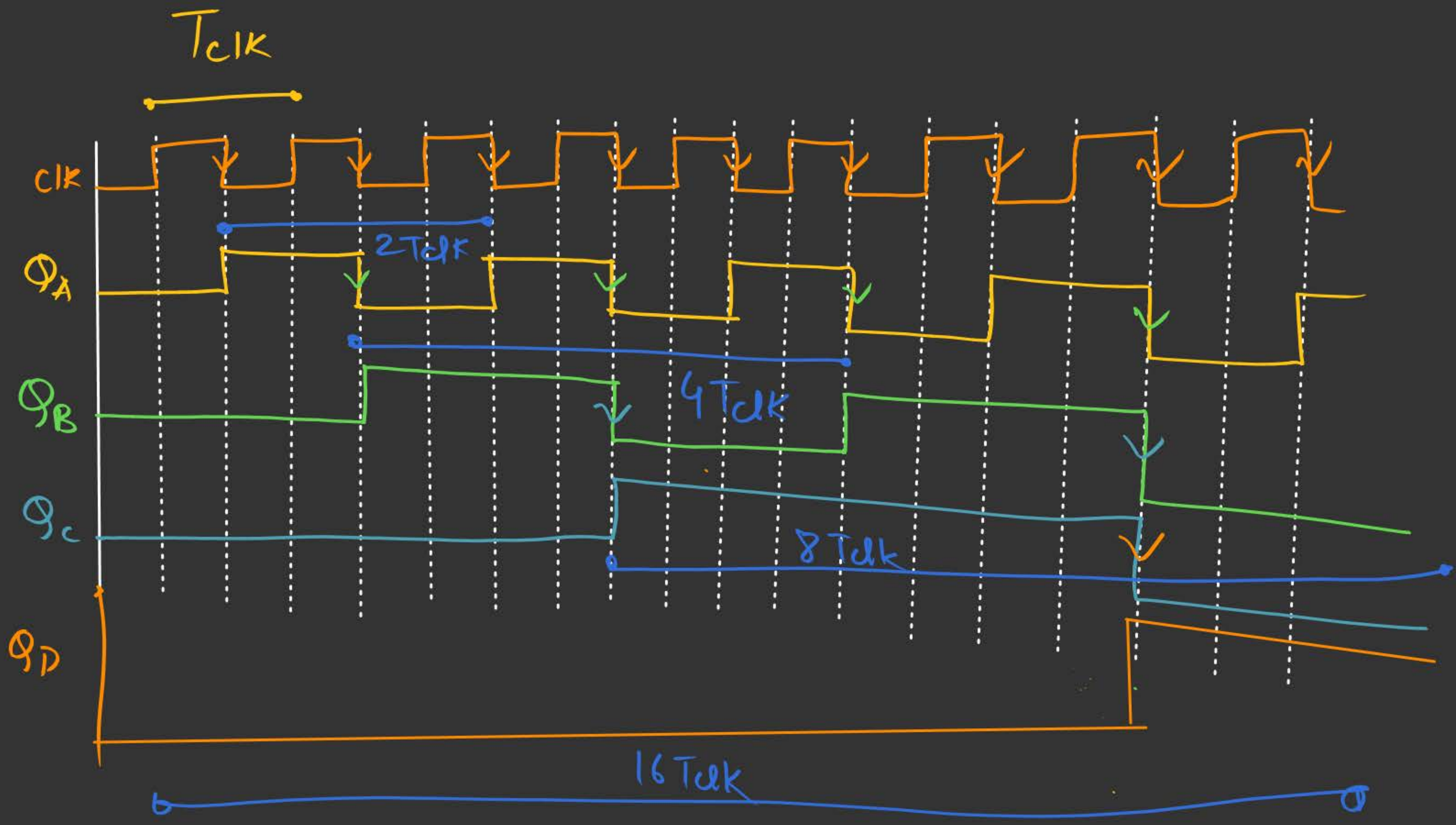


↳ -ve edge Trigg



$$\boxed{T_{clk} \geq 4 * T_{FF}}$$

(V.V. Imp)



$$F_{clk} = \frac{1}{T_{clk}}$$

$$T_{\phi A} = 2T_{clk} \Longrightarrow F_{\phi A} = \frac{1}{2T_{clk}} = \frac{F_{clk}}{2}$$

$$T_{\phi B} = 4T_{clk} \Longrightarrow F_{\phi_B} = \frac{1}{4T_{clk}} = \frac{F_{clk}}{4}$$

$$T_{\phi C} = 8T_{clk} \Longrightarrow F_{\phi_C} = \frac{F_{clk}}{8}$$

$$T_{\phi D} = 16T_{clk} \Longrightarrow F_{\phi_D} = \frac{F_{clk}}{16}$$

Imp Formulae :-

n-bit Asynchronous Counter

① $\boxed{T_{clk} \geq n * T_{PDFF}}$

frequency

$$\frac{1}{T_{clk}} \leq \frac{1}{n * T_{PDFF}}$$

② $\boxed{F_{clk} \leq \frac{1}{n * T_{PDFF}}}$ $\longrightarrow$ ③ $\boxed{F_{clk\,(max)} = \frac{1}{n * T_{PDFF}}}$

☆ Synchronous Counter:

- ① Ring
- ② Johnson

→ Here all the FFs are Synchronised with the Same clock.

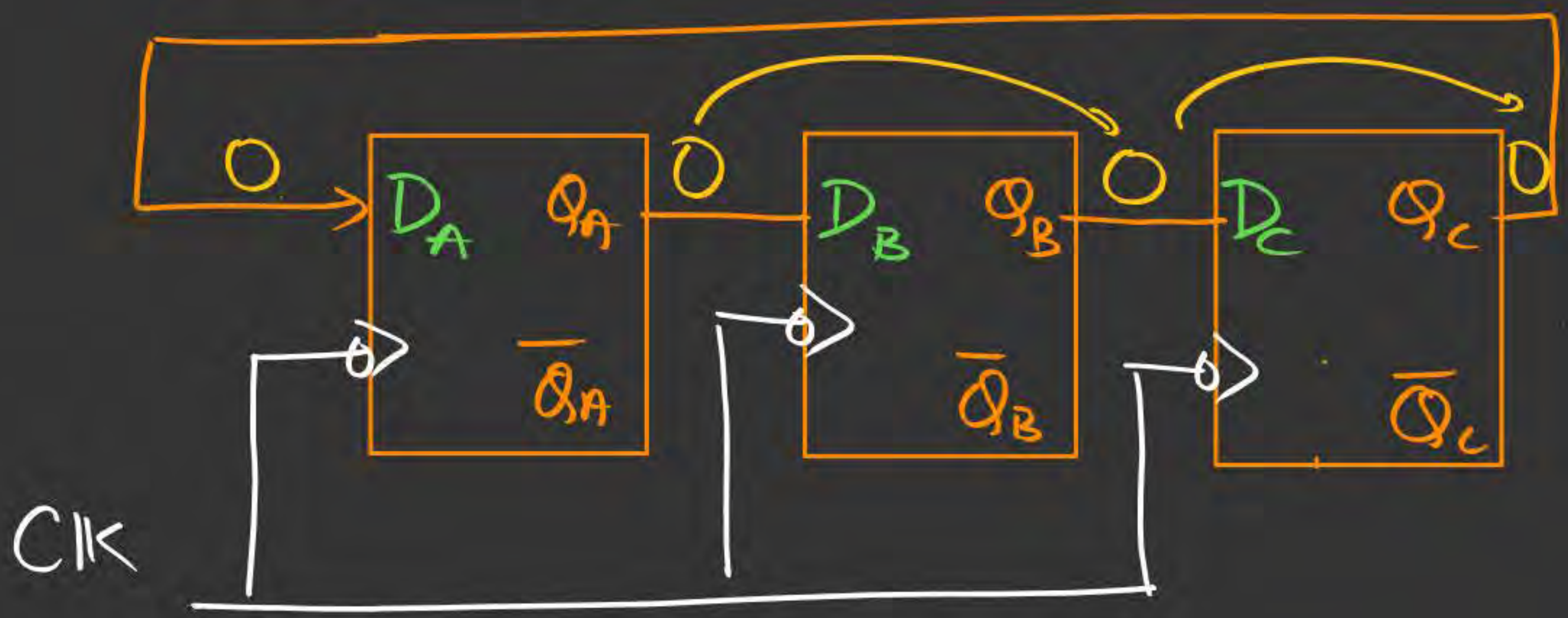
→ Default → Same Order as given

1 Ring Counter:

→ SISO Shift Register in the form of a Ring

1 Ring Counter

3-bit Ring Counter



Clock	Q_A	Q_B	Q_C	
0	0	0	0	
1	0	0	0	
2	0	0	0	
3	0	0	0	
4				
5				
6				
7				
8				

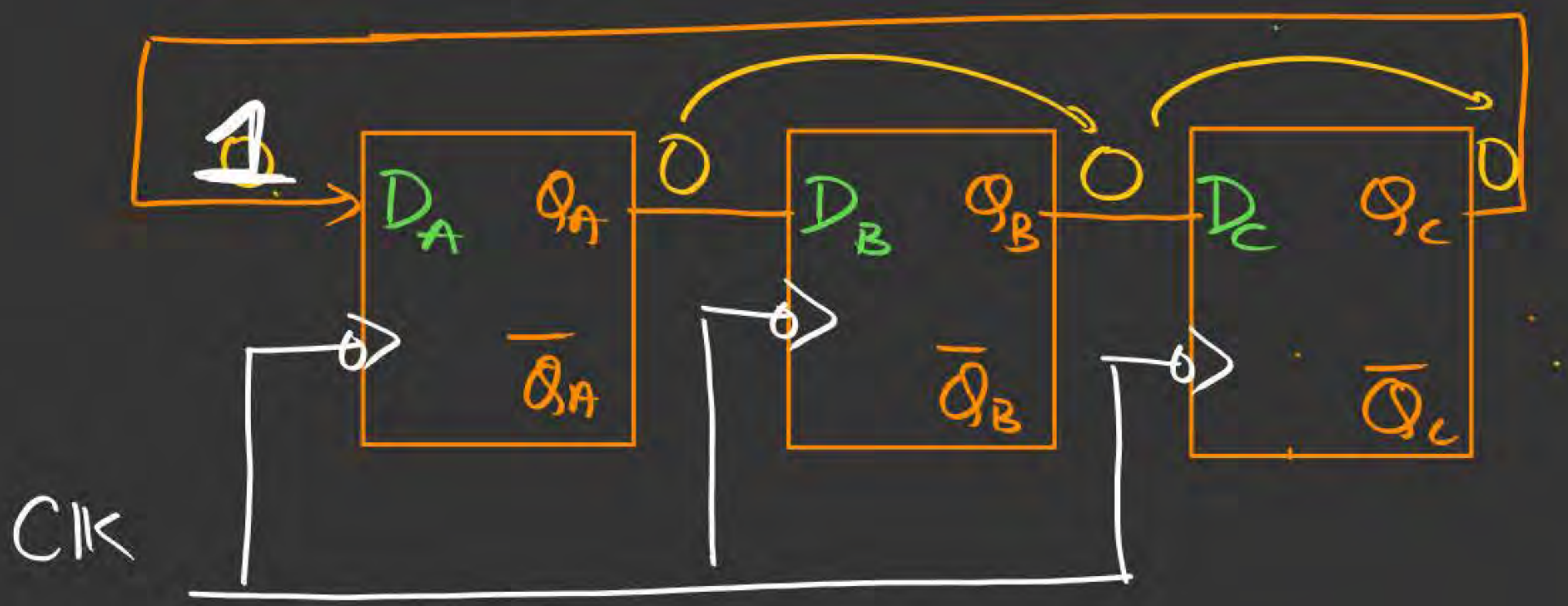
Imp :-

1) Ring Counter is not a self-starting counter. 1000

2) To start it, we have to place a '1' at the MSB along with the clock and then that 1 will rotate among all the Other FFs.

(1) Ring Counter

3-bit Ring Counter



Clock	Q_A	Q_B	Q_C	
0	0	0	0	
1	1	0	0	
2	0	1	0	
3	0	0	1	
4	1	0	0	
5				
6				
7				
8				

Observation :-

1) 3-bit Ring Counter

$$\underbrace{\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array}}_{\text{MOD-3}}$$

2) 4-bit Ring Counter.

$$\begin{array}{cccc} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array}$$

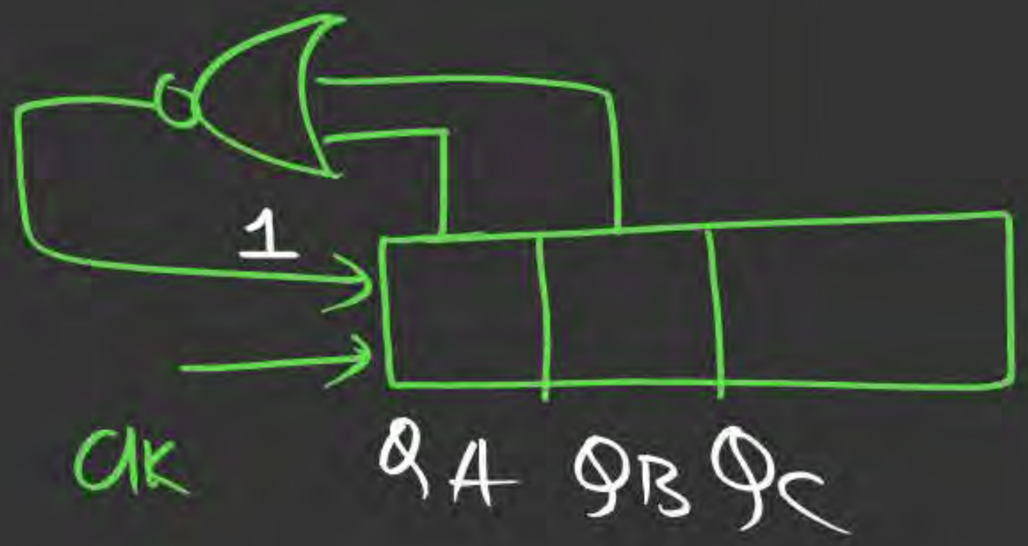
MOD-4

N-bit Ring Counter

$\Rightarrow$ MOD-N

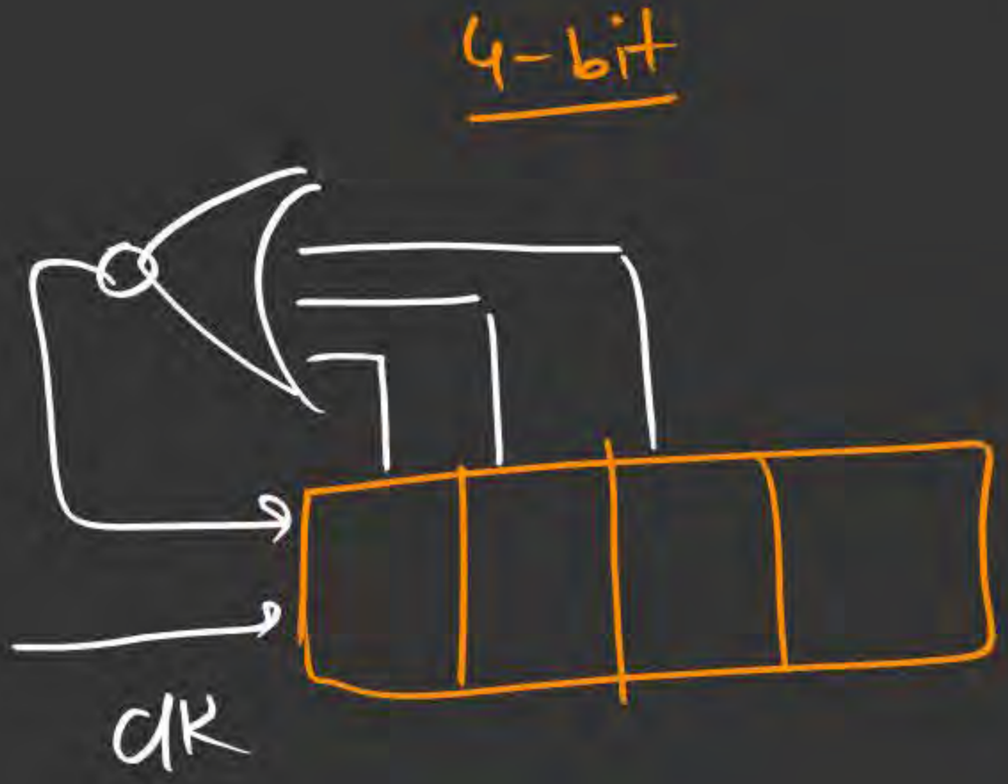
$$\boxed{\text{unused states} = 2^N - N}$$

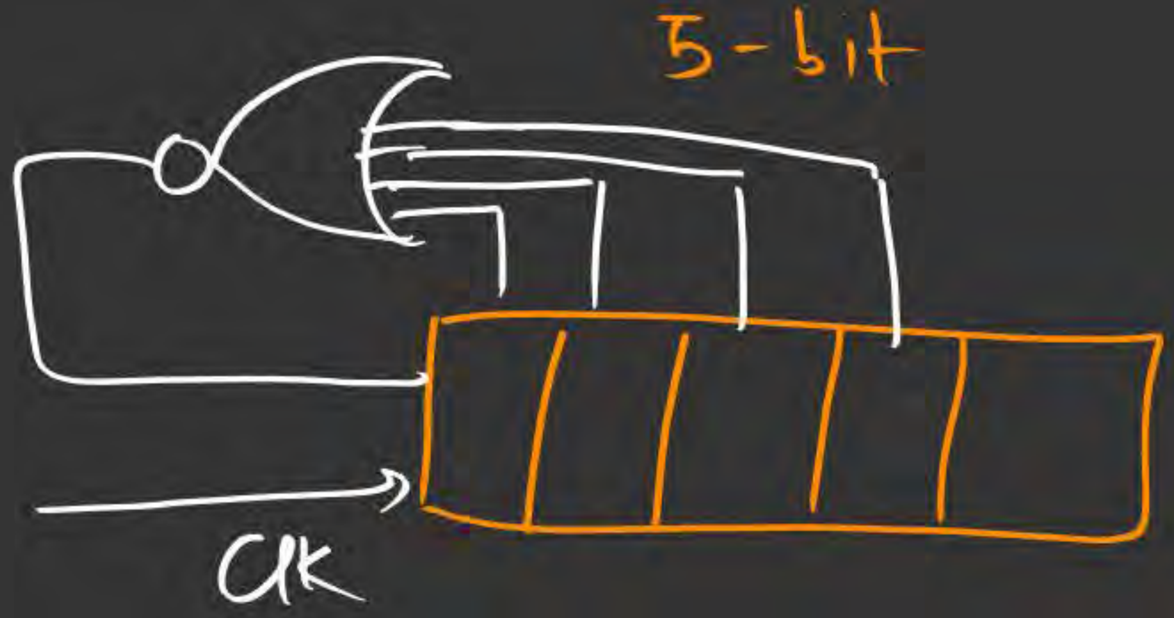
Self-starting Ring Counter



Clk	Q_A	Q_B	Q_C
0	0	0	0
1	1	0	0
2	0	1	0
3	0	0	1
4	1	0	0
5			

Self Starting Ring Counter





* Johnson Counter

Also known as :-

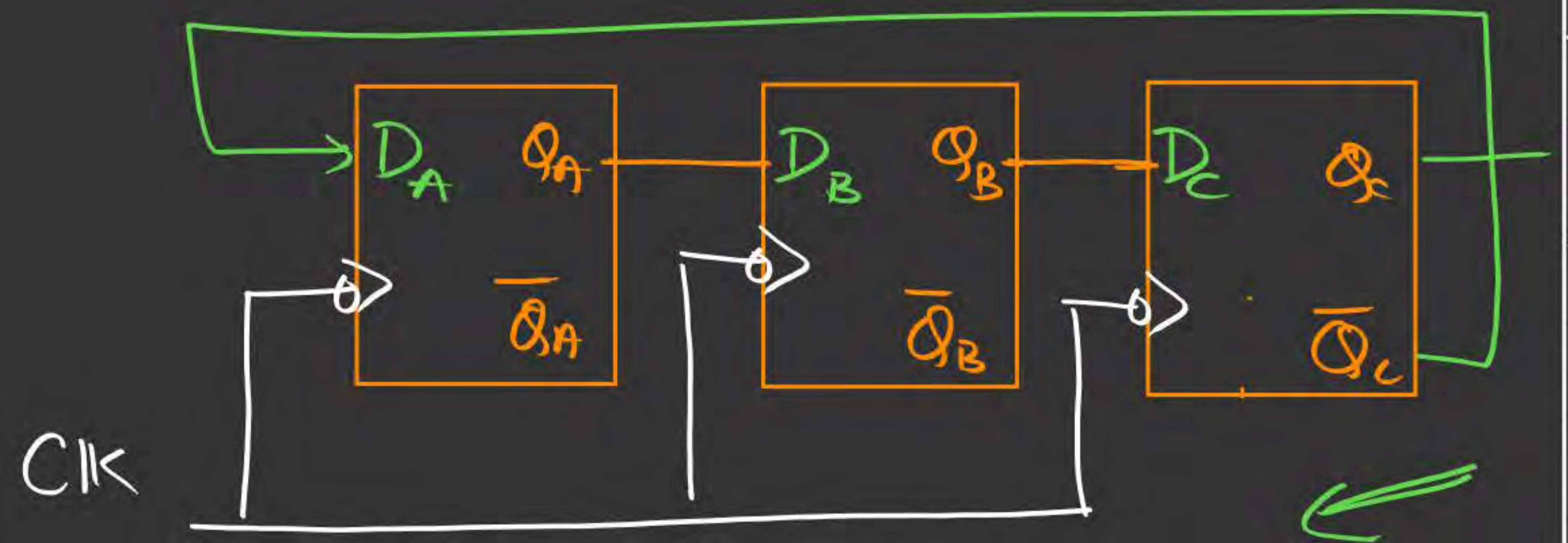
1> Creeping Counter

2> Walking "

3> Mobies "

$\boxed{\text{*4> Twisted Ring Counter}}$

Johnson Counter



Clock	Q_A	Q_B	Q_C	
0	0	0	0	
1	1	0	0	
2	1	1	0	
3	1	1	1	
4	0	1	1	
5	0	0	1	
6	0	0	0	
7	1	0	0	
8	1	1	0	

MOD-6

Shortcut: observation for Johnson Counter

1) 3-bit Johnson Counter:

000
100
110
111
011
001

$\underbrace{\quad}_{\text{mod-6}}$

unused: 101
010

2) 4-bit Johnson Counter

0000
1000
1100
1110
1111
0111
0011
0001

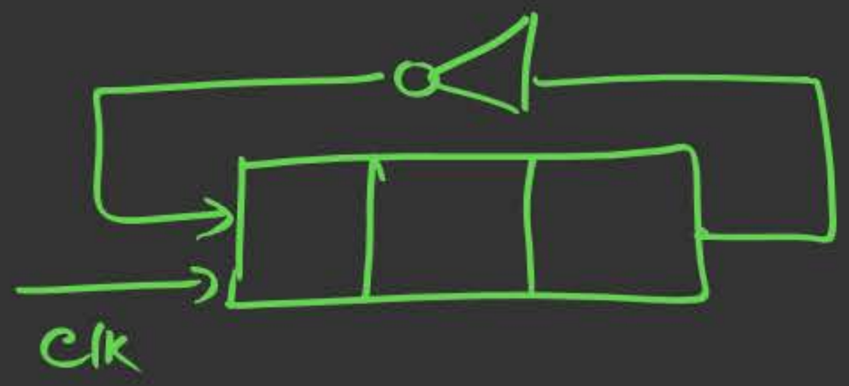
$\underbrace{\quad}_{\text{MOD-8}}$

V. Imp

N-bit Johnson Counter $= 2 * N$

Unused states $=$ Total $-$ MOD

$= \boxed{2^N - 2 * N}$



Clock	Q_A	Q_B	Q_C
0	1	0	1
1	0	1	0
2	1	0	1
3	0	1	0
4	1	0	1

Imp observations :

1) MOD-2

2) 101 ⇄ 010 (stuck between unused states)

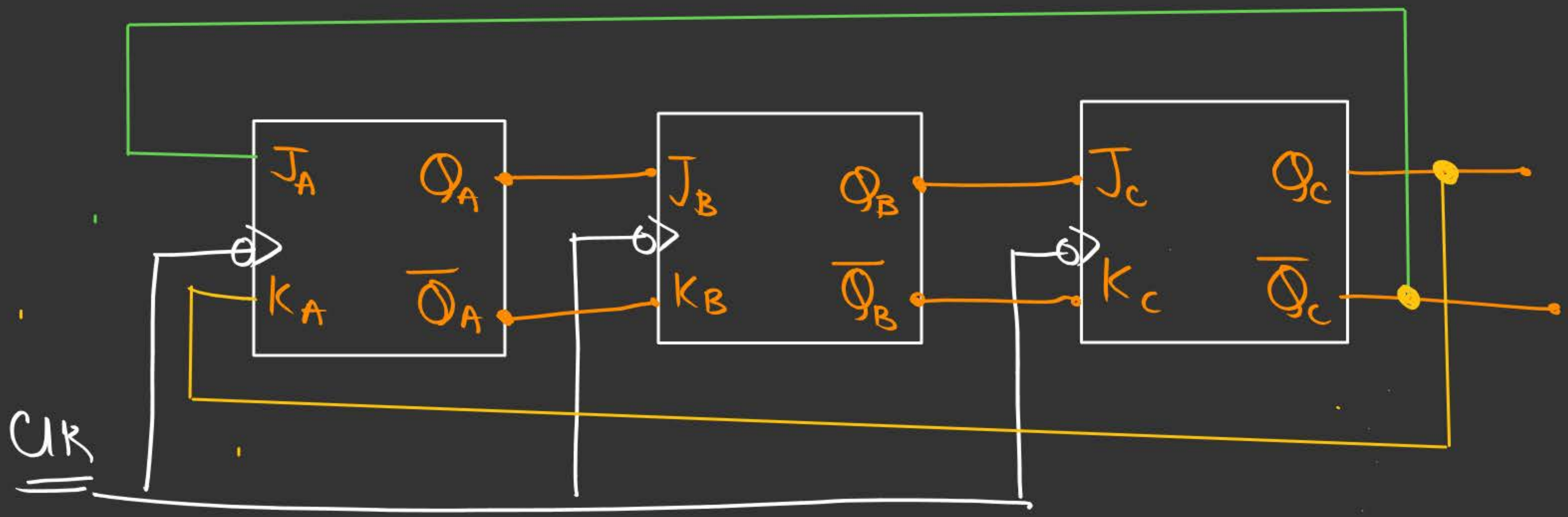
Imp :-

→ **Lockout Problem :-**

⇓

Whenever Johnson counter enters an unused state, it will be trapped / locked / stuck into the unused states, and this problem is called **Lockout Problem**.

Johnson Counter using JK ff



Designing a Synchronous Counter

Topic : DESIGNING OF SYNCHRONOUS COUNTER

STEP 1. Write the Previous and Present State.

STEP 2. Write the Excitation Table of FF.

STEP 3. Write the Logical Expression.

STEP 4. Minimize the Logical expression.

STEP 5. Hardware Implementation.

2 mins Summary

→ Counters → Async, Sync

Ring

Johnson

THANK - YOU